

## SAFETY DATA SHEET

Creation Date 29-April-2010

Revision Date 28-March-2024

Revision Number 3

### 1. Identification

**Product Name** Decafluorobiphenyl

**Cat No. :** L01152

**CAS-No** 434-90-2  
**Synonyms** Perfluorobiphenyl

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Specific target organ toxicity (single exposure)** Category 3  
Target Organs - Respiratory system.

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

May cause respiratory irritation

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER/ doctor if you feel unwell

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component          | CAS-No   | Weight % |
|--------------------|----------|----------|
| Decafluorobiphenyl | 434-90-2 | >95      |

### 4. First-aid measures

|                                                               |                                                                                                                 |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                                         | If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance.                 |
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention. |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.     |
| <b>Inhalation</b>                                             | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention if symptoms occur.           |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Get medical attention.                                                                  |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | No information available.<br>Treat symptomatically                                                              |

### 5. Fire-fighting measures

|                                                      |                                                                                       |
|------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                  | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>                | No information available                                                              |
| <b>Flash Point<br/>Method -</b>                      | No information available<br>No information available                                  |
| <b>Autoignition Temperature<br/>Explosion Limits</b> | No information available                                                              |

|                                  |                          |
|----------------------------------|--------------------------|
| Upper                            | No data available        |
| Lower                            | No data available        |
| Sensitivity to Mechanical Impact | No information available |
| Sensitivity to Static Discharge  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen fluoride.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

Health  
2

Flammability  
1

Instability  
0

Physical hazards  
N/A

**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Prevent product from entering drains.

**7. Handling and storage****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Pay attention to flashback.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep in properly labeled containers. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers recommendations |                 | Splash protection only |
| Neoprene       |                                   |                 |                        |

Natural rubber  
PVC

-

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

#### Environmental exposure controls

No information available.

#### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

## 9. Physical and chemical properties

|                                        |                              |
|----------------------------------------|------------------------------|
| Physical State                         | Crystalline Powder Solid     |
| Appearance                             | White                        |
| Odor                                   | No information available     |
| Odor Threshold                         | No information available     |
| pH                                     | Not applicable               |
| Melting Point/Range                    | 68 - 70 °C / 154.4 - 158 °F  |
| Boiling Point/Range                    | 206 °C / 402.8 °F @ 760 mmHg |
| Flash Point                            | No information available     |
| Evaporation Rate                       | Not applicable               |
| Flammability (solid,gas)               | No information available     |
| Flammability or explosive limits       |                              |
| Upper                                  | No data available            |
| Lower                                  | No data available            |
| Vapor Pressure                         | No information available     |
| Vapor Density                          | Not applicable               |
| Specific Gravity                       | No information available     |
| Solubility                             | No information available     |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | No information available     |
| Decomposition Temperature              | No information available     |
| Viscosity                              | Not applicable               |
| Molecular Formula                      | C12 F10                      |
| Molecular Weight                       | 334.12                       |

## 10. Stability and reactivity

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                 |
| Stability                        | Stable under recommended storage conditions.                               |
| Conditions to Avoid              | Incompatible products. Excess heat. Avoid dust formation.                  |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Strong bases                        |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen fluoride |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                   |

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information**

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** Irritating to respiratory system

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component          | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|----------|------------|------------|------------|------------|------------|
| Decafluorobiphenyl | 434-90-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the aquatic environment.

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

**UN-No**

UN3152

**Proper Shipping Name**

POLYHALOGENATED BIPHENYLS, SOLID

|                             |                                  |
|-----------------------------|----------------------------------|
| <b>Hazard Class</b>         | 9                                |
| <b>Packing Group</b>        | II                               |
| <b>TDG</b>                  |                                  |
| <b>UN-No</b>                | UN3152                           |
| <b>Proper Shipping Name</b> | POLYHALOGENATED BIPHENYLS, SOLID |
| <b>Hazard Class</b>         | 9                                |
| <b>Packing Group</b>        | II                               |
| <b>IATA</b>                 |                                  |
| <b>UN-No</b>                | UN3152                           |
| <b>Proper Shipping Name</b> | POLYHALOGENATED BIPHENYLS, SOLID |
| <b>Hazard Class</b>         | 9                                |
| <b>Packing Group</b>        | II                               |
| <b>IMDG/IMO</b>             |                                  |
| <b>UN-No</b>                | UN3152                           |
| <b>Proper Shipping Name</b> | POLYHALOGENATED BIPHENYLS, SOLID |
| <b>Hazard Class</b>         | 9                                |
| <b>Packing Group</b>        | II                               |

## 15. Regulatory information

### International Inventories

| Component          | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Decafluorobiphenyl | 434-90-2 | -   | -    | -    | -                                             | 207-107-3 | -      | -   |

| Component          | CAS-No   | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------|----------|-------|------|------|------|------|------|-------|-------|
| Decafluorobiphenyl | 434-90-2 | -     | -    | -    | X    | X    | -    | -     | -     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

**Authorisation/Restrictions according to EU REACH** Not applicable

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component          | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Decafluorobiphenyl | 434-90-2 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No | Seveso III Directive (2012/18/EC) - | Seveso III Directive (2012/18/EC) - | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|

|                    |          | Qualifying Quantities<br>for Major Accident<br>Notification | Qualifying Quantities<br>for Safety Report<br>Requirements |                |                |
|--------------------|----------|-------------------------------------------------------------|------------------------------------------------------------|----------------|----------------|
| Decafluorobiphenyl | 434-90-2 | Not applicable                                              | Not applicable                                             | Not applicable | Not applicable |

## 16. Other information

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28-March-2024

**Print Date**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**